PURITY OF BRANCH LOCUS (NAGATA): A QUASI-FINITE MORPHISM F : X -> Y OF LOCALLY NOETHERIAN SCHEMES, WITH O_{X,X} NORMAL, O_{Y,Y} REGULAR, EQUAL RELATIVE DIMENSIONS AT LEAST 1 AT X, AND F UNRAMIFIED IN CODIMENSION 1 AROUND X, IS ETALE AT X.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5334-5365 of sources/stacks-pione/pinned-original-source.tex; proof span is lines 5367-5430, the complete author proof (62 body lines).

Editorial note. Navigation stubs: 11 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 16 of queue_pione.json (pb=2947, stmt 5334-5365, proof 5367-5430).

Editorial note. pione wave-1: fresh-adjudication accept (2026-10-14); label not covered by the 85 old-session S3-pi cards.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00PD: sources/stacks-pione/pinned-original-source.tex, source lines 5430–5430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00ON: sources/stacks-pione/pinned-original-source.tex, source lines 5430–5430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AFS: sources/stacks-pione/pinned-original-source.tex, source lines 5430–5430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0567: sources/stacks-pione/pinned-original-source.tex, source lines 5430–5430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:pione.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00H8: sources/stacks-pione/pinned-original-source.tex, source lines 5430–5430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02VN: sources/stacks-pione/pinned-original-source.tex, source lines 5430–5430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 036W: sources/stacks-pione/pinned-original-source.tex, source lines 5430–5430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 025L: sources/stacks-pione/pinned-original-source.tex, source lines 5430–5430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BMA: sources/stacks-pione/pinned-original-source.tex, source lines 5284–5292. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AGY: sources/stacks-pione/pinned-original-source.tex, source lines 5430–5430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02LK: sources/stacks-pione/pinned-original-source.tex, source lines 5430–5430. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1 (Purity of branch locus)**.** *and This result was first stated and proved by Zariski in geometric form in . The generalization to nonperfect ground fields by Nagata was published as the next article in the same volume of the Proceedings of the National Academy of Sciences of the United States of America in . In the following year Nagata proved the result for Noetherian local rings in . His proof uses a result of Chow which is a Bertini theorem for complete local domains, see ; the history of Bertini's theorems is discussed in Kleiman's historical article . A few years later a completely different proof was found by Auslander, see . Let $f : X \to Y$ be a morphism of locally Noetherian schemes. Let $x \in X$ and set $y = f(x)$. Assume*

(1) $\mathcal{O}_{X,x}$ *is normal,*
(2) $\mathcal{O}_{Y,y}$ *is regular,*
(3) f *is quasi-finite at* x*,*
(4) $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) \geq 1$
(5) *for specializations* $x' \rightsquigarrow x$ *with* $\dim(\mathcal{O}_{X,x'}) = 1$ *our* f *is unramified at* x'*.*

Then f is étale at x.

Proof. We will prove the lemma by induction on $d = \dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y})$.

An uninteresting case is when $d = 1$. In that case we are assuming that f is unramified at x and that $\mathcal{O}_{Y,y}$ is a discrete valuation ring (Algebra, Lemma 00PD). Then $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,y}$ (otherwise the map would not be quasi-finite at x) and

we see that f is flat at x. Since flat + unramified is étale we conclude (some details omitted).

The case $d \geq 2$. We will use induction on d to reduce to the case discussed in Lemma 0BMA. To check f is étale at x we may work étale locally on the base and on the target (Descent, Lemmas 02VN and 036W). Thus we can apply More on Morphisms, Lemma 02LK and assume that $f : X \to Y$ is finite and that x is the unique point of X lying over y. Here we use that étale extensions of local rings do not change dimension, normality, and regularity, see More on Algebra, Section 0AGY and Étale Morphisms, Section 025L.

Next, we can base change by $\mathrm{Spec}(\mathcal{O}_{Y,y})$ and assume that Y is the spectrum of a regular local ring. It follows that $X = \mathrm{Spec}(\mathcal{O}_{X,x})$ as every point of X necessarily specializes to x.

The ring map $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ is finite and necessarily injective (by equality of dimensions). We conclude we have going down for $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ by Algebra, Proposition 00H8 (and the fact that a regular ring is a normal ring by Algebra, Lemma 0567). Pick $x' \in X$, $x' \neq x$ with image $y' = f(x')$. Then $\mathcal{O}_{X,x'}$ is normal as a localization of a normal domain. Similarly, $\mathcal{O}_{Y,y'}$ is regular (see Algebra, Lemma 0AFS). We have $\dim(\mathcal{O}_{X,x'}) = \dim(\mathcal{O}_{Y,y'})$ by Algebra, Lemma 00ON (we checked going down above). Of course these dimensions are strictly less than d as $x' \neq x$ and by induction on d we conclude that f is étale at x'.

Thus we arrive at the following situation: We have a finite local homomorphism $A \to B$ of Noetherian local rings of dimension $d \geq 2$, with A regular, B normal, which induces a finite étale morphism $V \to U$ on punctured spectra. Our goal is to show that $A \to B$ is étale. This follows from Lemma 0BMA and the proof is complete. $\square$